📘 MODULE 30: SPLUNK ENTERPRISE SECURITY (ES)

30.1 What ES is

Splunk Enterprise Security is a **premium, separately licensed app** installed on a **search head** that turns Splunk into a SIEM. It adds: a framework for **detections**, **triage workflow**, **risk-based alerting**, **asset/identity context**, and **threat intelligence**.

	Splunk Enterprise	Splunk ES
What	Core platform	App on top
Gives you	Search, dashboards, alerts	Notables, incident review, risk, assets/identities, threat intel, security content
License	Core ingest license	Extra paid license

Version note: ES has changed considerably across major versions (for example, newer releases revamp the analyst interface and rename some concepts such as notable events). The ideas below are stable, but **menu names and terms vary by version**. Check your ES version's documentation, and look at what your target employer uses.

Lab note: ES is not in the free Enterprise download. Splunk has offered trials and hosted sandboxes, so check Splunk's site for current availability. For free hands-on, you can practice the same ideas with the **Boss of the SOC (BOTS)** datasets published by Splunk on GitHub (verify the current repo and instructions).

30.2 Core concepts

Notable events

A **notable event** is a work item created when a correlation search matches. It appears in **Incident Review**, with a title, urgency, status, and owner.

Correlation searches

Scheduled searches (Module 29) that create notables and/or risk.

Incident Review (the L1 workspace)

The main triage dashboard:

- A list of notables with **urgency**, **status**, **owner**, **time**.
- Filters (urgency, status, owner, security domain).
- **Notable details:** fields from the search, the **drilldown** link to raw events, history, and related notables.

Urgency is computed from the detection's **severity** and the **priority** of the affected asset/identity (a critical server makes the same alert more urgent).

Notable status workflow (L1 typical)

```
New → In Progress → Resolved → Closed
        (assigned to analyst)  (with disposition/comment)
```

Statuses are configurable.

Assets and identities

- **Asset framework:** a table of systems (IP, MAC, DNS name, owner, **priority**, category). ES uses it to enrich fields like `src` and `dest`.
- **Identity framework:** a table of users (name, email, department, priority).
- **Why it matters:** the analyst sees "src=10.0.0.5 → `HR-Laptop-12`, priority=high" instead of a bare IP. This is the same idea as a lookup (Module 11).

Risk framework / Risk-Based Alerting (RBA)

Instead of alerting on every small signal:

1. Detections add **risk modifiers**: a **risk score** against a **risk object** (a user, system, or other entity), with MITRE ATT&CK tags where available.
2. Scores **accumulate** per risk object in the Risk index.
3. A **risk threshold correlation search** creates a **risk notable** when an entity's total risk crosses a limit (or multiple distinct techniques are seen).

Why: fewer, higher-quality alerts. 20 low-fidelity events against one user are more convincing than one noisy alert.

Threat intelligence framework

ES ingests **threat feeds** (IPs, domains, hashes, URLs) into collections, and **threat match** searches compare them with your data (via CIM data models). A match produces a notable. Feeds and sources are configured in ES.

Investigations

A shared workspace for a case: you add **notes, notables, events, searches, and artifacts** and a timeline, and collaborate with others. In newer versions this is merged into broader case-style workflows (verify your version).

Adaptive Response

Response actions that run from a correlation search or from Incident Review, for example: add to a watchlist, ping a host, run a script, create a ticket, send a notification, or call SOAR (Module 31). These can be automatic or manual.

30.3 Typical L1 workflow in ES

1. Open **Incident Review**, filter by **urgency**, **status=New**, and your security domain.
2. **Assign** the notable to yourself (status → In Progress).
3. Read the **description, fields, risk score, and related notables**.
4. Use the **drilldown** to see underlying events and check context (asset, identity, threat intel).
5. Decide **true positive vs false positive** (using your playbook).

6. Add **comments**, then **escalate to L2** or **close** with a disposition (e.g. false positive, resolved).
7. Document in the ticket system.

30.4 Typical L2 work

- Deeper investigation with ad-hoc SPL, pivots, and `tstats`.
- **Tuning** correlation searches: thresholds, allowlists, throttling.
- Building and testing **new detections** (against CIM).
- Checking **data onboarding** and CIM mapping quality.
- Maintaining the asset/identity tables.

30.5 What ES searches look like (example)

spl

```
| tstats summariesonly=true count, values(Authentication.user) AS users
  FROM datamodel=Authentication
  WHERE Authentication.action="failure"
  BY Authentication.src
| rename Authentication.* AS *
| where count>=20
```

In ES, this logic is saved as a **correlation search** with a **notable** action and the fields to show.

30.6 ES vs plain Splunk (do not mix them)

Need	Plain Enterprise	ES
Run SPL, dashboards, alerts	Yes	Yes
Triage workflow (assign, status)	No (build it yourself)	Incident Review
Risk scoring	Manual	Risk framework
Asset/identity enrichment	Manual lookups	Built-in framework
Threat intel matching	Manual lookups	Threat intel framework
Requires CIM	No	Yes, practically required

30.7 Hands-on (no ES license needed)

1. Write the Module 29 detection with CIM `tstats`, and verify your data feeds the Authentication model.
2. Build a "mini Incident Review" with a lookup: store alerts in `incidents.csv` using `outputlookup append=true` (columns: time, src_ip, status, owner), and display them in a dashboard table.
3. Create `assets.csv` (ip, hostname, priority), then enrich your detection with `lookup` and sort by priority. This is the same idea as ES asset enrichment.
4. Practice explaining aloud: "What happens from log to closed ticket?"

30.8 Troubleshooting ES (concepts)

Symptom	Cause
No notables	Correlation search disabled, data model not accelerated or not populated, threshold too high
Notables missing context	Asset/identity lists empty or not matching
Searches slow	Data models not accelerated, or too many correlation searches at once
Data not showing in ES dashboards	Not CIM-mapped, or index not in the CIM/ES allowed indexes
Too many notables	Needs tuning, throttling, or RBA

30.9 Interview Q&A

- **What is Splunk ES?** A premium SIEM app on Splunk that provides correlation searches, notables, incident review, risk-based alerting, asset/identity and threat-intel enrichment.
- **What is a notable event?** A work item created by a correlation search that analysts triage in Incident Review.
- **What is risk-based alerting?** Detections add risk scores to entities, and an alert fires when an entity's cumulative risk crosses a threshold, which reduces noise.
- **How is urgency determined?** From the detection's severity and the priority of the involved asset/identity.
- **How does ES depend on CIM?** Its searches and dashboards run on CIM data models, so data must be mapped.
- **How would you investigate a notable?** Review the context, use the drilldown to see the raw events, check asset/identity/threat intel, decide TP/FP, document, then escalate or close.
- **ES vs Enterprise?** ES is a paid app layered on Enterprise that adds SIEM workflow.

30.10 Revision

ES = paid SIEM app · correlation search → notable → Incident Review · urgency = severity × priority · risk = accumulating scores on entities · assets/identities enrich · needs CIM · terms vary by version.

📘 MODULE 36: SPLUNK TROUBLESHOOTING

36.0 The master workflow

Always follow the data path from **source to search**. Find where it stops.

```
1. Is the SOURCE producing data?          (file growing? app logging?)
2. Is the INPUT configured + readable?    (inputs.conf, permissions)
3. Is the FORWARDER connected?            (outputs.conf, port 9997)
4. Did the INDEXER RECEIVE it?            (_internal metrics, tcpin)
5. Is it in the RIGHT INDEX?              (index exists, spelling)
6. Is the TIME right?                     (timestamp, time range)
7. Can I SEARCH it?                       (role permissions, index allowed)
8. Are FIELDS extracted?                  (sourcetype, props, permissions)
```

Top tools: `splunk btool`, `index=_internal`, `splunk list forward-server`, the Job Inspector, `ss -tlnp`, `ls -l`.

36.1 "No events found" (most important)

SYMPTOM: search returns "No results found."
POSSIBLE CAUSES: wrong time range; wrong index name; role can't search that index; data never arrived; source isn't producing; typo or wrong case in a field value.
COMMANDS/SPL:

```spl
| tstats count WHERE index=* BY index
| metadata type=sourcetypes index=*
index=* sourcetype=<mine> earliest=0
```

CONFIG CHECK: `splunk btool inputs list --debug`; Settings → Indexes; role's allowed indexes.
FIX: widen the time to All time, correct the index or sourcetype, grant the role access, or fix the input.
VERIFY: the search returns events, and `_time` looks right.

36.2 "Index does not exist"

SYMPTOM: the input or search refers to an index and the data is missing; `splunkd.log` has a warning.
CAUSES: the index was never created on the indexer; typo; it was created on a search head only; or a forwarder points to an index that exists only on another instance.
SPL:

```spl
index=_internal source=*splunkd.log* "unconfigured/disabled/deleted index"
| rest /services/data/indexes splunk_server=local | table title disabled
```

FIX: create the index **on the indexers** (`indexes.conf` or UI) and restart if needed.
VERIFY: new events appear. Events already dropped are lost, so re-ingest if possible.

36.3 "Data is in the wrong index"

CAUSES: missing `index =` in the input (defaults to `main`); another stanza overriding it; a transform routing it.
CHECK: `splunk btool inputs list monitor --debug`

```spl
index=* source="/var/log/auth.log" | stats count by index
```

FIX: set `index` in the correct stanza, restart, and note that **old events stay where they are** (copying or re-indexing is needed to move them).

36.4 "Wrong sourcetype"

SYMPTOM: events broken wrongly, odd fields, no timestamps.
CAUSES: automatic sourcetype guessing; no explicit sourcetype; a rule in `props.conf` overriding.

```spl
index=* source="<path>" | stats count by sourcetype
```

CHECK: `btool inputs list --debug` and `btool props list <sourcetype> --debug`.
FIX: set `sourcetype =` explicitly in `inputs.conf`, and restart. Only new events are affected.

36.5 "Fields are missing"

CAUSES: wrong sourcetype; no extraction defined; the regex doesn't match; extraction is private/not shared; the field exists only in some events; the add-on is not installed on the search head.
CHECK:

```spl
index=<idx> sourcetype=<st> | fieldsummary maxvals=5 | table field count distinct_count
```

Check the search mode (**Verbose** shows all fields, while "Fast" mode hides some). Settings → Fields → Field extractions → permissions.
FIX: correct the sourcetype, install the add-on on the **search head**, add an extraction (`EXTRACT-` in props), and share it.

36.6 "Timestamp is incorrect"

SYMPTOM: events appear hours off, in the future, or at index time.
CAUSES: time zone mismatch; wrong `TIME_FORMAT` or `TIME_PREFIX`; Splunk found a different date in the text; no timestamp in the event (uses time received).

```spl
index=<idx> | eval lag=_indextime-_time | stats avg(lag) max(lag) by sourcetype
index=_internal component=DateParserVerbose
```

CONFIG:

```ini
[my_app_logs]
TIME_PREFIX = ^\[
TIME_FORMAT = %Y-%m-%d %H:%M:%S
MAX_TIMESTAMP_LOOKAHEAD = 19
TZ = Asia/Kolkata
```

(use the time zone of the log's source; `Asia/Kolkata` is only an example)
FIX: place on the **indexer/HF (parsing tier)**, restart. **Only new events** are corrected.
VERIFY: `lag` is near zero.

36.7 "Forwarder is not sending"

CAUSES: UF stopped; `outputs.conf` wrong; receiver closed; firewall; UF can't read the file; no new data in the file.
On the forwarder (Linux):

```bash
sudo /opt/splunkforwarder/bin/splunk status
sudo /opt/splunkforwarder/bin/splunk list forward-server
sudo /opt/splunkforwarder/bin/splunk btool outputs list --debug
tail -50 /opt/splunkforwarder/var/log/splunk/splunkd.log
```

`list forward-server` shows **Active** vs **Configured but inactive** forwards.
On the indexer:

```spl
index=_internal source=*metrics.log* group=tcpin_connections
| stats latest(_time) AS last_seen by hostname, sourceIp
```

FIX: start the UF, correct `server=` in `outputs.conf`, restart, and open the network path.
VERIFY: the host appears in the search above and events arrive.

36.8 "Receiver is not accepting" and 36.9 "Port 9997 problem"

SYMPTOM: UF log: "Connection refused," "Cannot connect to…," or "Connected but no data."
CAUSES: receiving port not enabled on the indexer; the port is enabled but a **firewall blocks it** (the port can be "open" in a sense but not for your source); SSL mismatch; the indexer is down or busy (queues blocked); wrong IP/DNS.
COMMANDS:

```bash
```

```
ss -tlnp | grep 9997                    # on indexer: is Splunk listening?
nc -zv <indexer-ip> 9997                # on forwarder: is the port reachable?
sudo ufw status                         # Ubuntu firewall
sudo /opt/splunk/bin/splunk display listen -auth admin:<pw>
```

spl

```
index=_internal source=*splunkd.log* component=TcpInputProc
index=_internal source=*metrics.log* group=queue blocked=true
```

FIX: `splunk enable listen 9997`, open the firewall, correct IP/port, fix SSL settings, and clear indexer queue problems.
Key idea: "port reachable" ≠ "data flowing." Check the UF's `splunkd.log`, the indexer's `tcpin_connections`, and queues.

36.10 "HEC problem"

CAUSES: HEC not enabled globally; token disabled; **index not allowed for the token**; wrong token or header; wrong endpoint or JSON; HTTPS vs HTTP mismatch (HEC uses SSL by default).
TEST:

bash

```
curl -k https://localhost:8088/services/collector/health
curl -k https://localhost:8088/services/collector/event \
  -H "Authorization: Splunk <token>" \
  -d '{"event":"hello","index":"main","sourcetype":"hec_test"}'
```

- `-k` skips certificate checking (lab only). `-H` sets the header.
- Success returns `{"text":"Success","code":0}`. Error responses name the problem (bad token, disabled, invalid index, bad JSON).
  FIX: Settings → Data inputs → HTTP Event Collector → enable Global Settings, check the token's **allowed indexes** and default index, fix the URL/token.

36.11 "Search is slow"

CAUSES: broad base search (`index=*`, All time); leading wildcards; `join` / `transaction` / `map`; many unneeded fields; high-cardinality `by`; too many concurrent searches.
TOOLS: Job → Inspect Job (execution costs: which phase took the time), and search mode **Fast**.
FIX:

- Narrow `index`, `sourcetype`, and time first.
- `fields` early to drop unneeded fields.
- Replace `join` / `transaction` with `stats`.
- Use `tstats`/accelerated data models for counts.
- Use summary indexing/scheduled searches for heavy dashboards.
  VERIFY: compare the job inspector before and after.

36.12 "Lookup not working"

CAUSES: file not uploaded or definition missing; not shared; field name mismatch; case/whitespace; automatic lookup is bound to the wrong sourcetype.
CHECK:

```spl
| inputlookup users_lookup
```

Then compare field values: `| eval check=len(user)` to spot hidden spaces.
FIX: share permissions, add `AS`, normalize with `lower()` / `trim()`, or set `case_sensitive_match=false` on the definition.

36.13 "Regex not extracting"

CHECK: test with `| head 20 | rex ... | table _raw field`; escape special characters; make variable parts optional; check for a missing `?<name>`; check `field=`.

```spl
... | rex "user=(?<user>\w+)" | where isnull(user)
```

This shows the events your regex fails on.
FIX: adjust the regex for the variants, then make it permanent via props or the Field Extractor and share it.

36.14 "Props/transforms not working"

CAUSES: wrong **tier** (parsing settings on a UF, or on a search head when the data goes through a different indexer); the stanza name doesn't match the sourcetype/source/host; no restart; **the transform name in props doesn't match** `transforms.conf`; regex mismatch; wrong `DEST_KEY`.
CHECK:

```bash
sudo /opt/splunk/bin/splunk btool props list <sourcetype> --debug
sudo /opt/splunk/bin/splunk btool transforms list <transform_name> --debug
sudo /opt/splunk/bin/splunk btool check
```

FIX: put the config on the first full Splunk instance in the path (indexer or HF), restart, and test with **new** events.
Remember: index-time changes don't change old data.

36.15 "Configuration change not taking effect"

Checklist:

1. Did you edit `local/`, not `default/`? (btool shows the winner)
2. Right **app** and **instance** (UF vs HF vs indexer vs search head)?
3. Did you **restart** (index-time) or refresh (search-time)?
4. Typo in stanza or attribute? (`btool check`)

5. Overridden by a higher-precedence file? (`btool --debug`)
6. Testing with **old** events? (index-time changes affect only new ones)

36.16 "Permission denied"

Two different types, and you must separate them:

- **OS-level:** Splunk can't read a file.

```bash
ls -l /var/log/auth.log
sudo -u splunk head -1 /var/log/auth.log
```

Fix with group membership (for example `adm`), file ACLs (`setfacl`), or log permissions, and restart.

```spl
index=_internal source=*splunkd.log* "Permission denied"
```

- **Splunk-level:** a user can't search an index or open an object. Check **Settings** → **Roles** (allowed indexes, capabilities) and the object's **Permissions** (Private/App/All apps).

36.17 "Dashboard panel not loading"

CAUSES: search error or timeout; token not set; wrong token name; invalid XML; the underlying report/lookup is private; the viewer's role lacks index access; heavy search.
CHECK: click the panel's **Open in Search**, and run it with the token values replaced. Check the browser console and **Edit** → **Source** for XML errors.
FIX: set token defaults, share the objects, optimize or accelerate the search.

36.18 "Alert not triggering"

See **Module 28.10**. Quick path: run the search manually with the same time range → check the schedule → `index=_internal sourcetype=scheduler savedsearch_name="..."` for `status` → trigger condition → throttle → action settings (email).

36.19 Troubleshooting Splunk itself: key logs

File (`/opt/splunk/var/log/splunk/`)	Use
`splunkd.log`	Main log (errors and warnings)
`metrics.log`	Throughput, queues, connections
`scheduler.log`	Scheduled searches/alerts
`splunkd_access.log`	Splunk Web/REST requests
`license_usage.log`	License use

```spl
index=_internal source=*splunkd.log* (log_level=ERROR OR log_level=WARN) | stats count by
```

36.20 Practice scenarios (troubleshoot before reading the fix)

Answer each with your **first 4 checks** (don't peek).

1. "My UF is running but Splunk has no events."
2. "`index=main` works but `index=web` gives nothing."
3. "My search returns events but `user` is missing."
4. "Events show tomorrow's date."
5. "My alert runs but no email arrives."
6. "A lookup works in search but not in the dashboard."

36.21 Interview Q&A

- **"No data from a forwarder, what do you do?"** Walk the path: source → input → forwarder status/outputs → network/port → indexer receiving and `tcpin_connections` → correct index → time range and permissions.
- **How do you find the effective config?** `splunk btool <file> list --debug`.
- **Where do you look for Splunk's own errors?** `index=_internal` (`splunkd.log`, `metrics.log`, `scheduler.log`).
- **Wrong timestamps?** Check `TIME_FORMAT`, `TIME_PREFIX`, `TZ` in props, on the parsing tier. Only new data is fixed.
- **A search is slow, what do you do?** Job inspector, narrow the scope, filter early, replace `join`/`transaction`, use `tstats` or accelerated data models.
- **Port 9997 is open but no data?** Verify the listener, firewall, `outputs.conf`, UF log, indexer `tcpin` metrics, queues, and index existence.

36.22 Revision

Follow the data path · `_internal` is your friend · `btool --debug` finds the winning config · index-time changes only affect new data · put parsing config on the right tier · permissions are both OS-level and Splunk-level.